

Texture Resonance Retrieval for Audited Guitar-Effect Preset Selection

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Abstract—Digital audio workstations expose rich effect chains, but mapping perceptual intent to editable low-level parameters remains ill-posed. We study retrieval-grounded guitar-effect preset selection, where the system returns an executable plugin configuration rather than a finalized waveform. Our method, Texture Resonance Retrieval (TRR), represents audio with Gram matrices of projected mid-level Wav2Vec2 activations to capture texture-relevant co-activation structure. On an audited 204-query guitar-effect benchmark with 1,063 candidate presets, TRR improves retrieval over selected audio embedding baselines, while mechanism ablations and near-duplicate sensitivity analyses define the robustness boundary of this result. A supervised regression boundary, listening study, and local vLLM diagnostic further show that parameter-space alignment should not be treated as a direct proxy for human perceptual preference.

Index Terms—guitar-effect preset retrieval, audio retrieval, digital audio workstations, texture representation, interpretable audio systems

I. INTRODUCTION

The field of neural audio synthesis currently faces a fundamental tension between *fidelity* and *control*. Recent high-fidelity generative models, primarily based on latent diffusion [1, 2], can achieve strong perceptual quality but often operate as “black boxes.” They generate finalized waveforms where internal attributes—such as the attack time of a compressor or the decay of a reverb—are entangled and inaccessible for post-hoc editing. Conversely, Differentiable Digital Signal Processing (DDSP) [3] offers interpretable, structured control but often suffers from the ill-posed nature of the inverse parameter estimation problem.

This tension motivates the development of retrieval-based audio control systems that can map perceptual intent to interpretable DSP parameters. By retrieving relevant presets rather than generating parameters from scratch, such systems are designed to produce editable outputs that integrate with DAW workflows. In this view, retrieval is valuable not because it solves the inverse problem in closed form, but because it places the query in a plausible executable neighborhood of parameter space that the user can still inspect and refine.

The key challenge is grounding perceptual descriptions in parameter space via retrieval. Standard audio representations (e.g., CLAP-style embeddings or mean-pooled Wav2Vec2 features) summarize an audio segment into a single vector; while effective for coarse semantics, such pooling can discard temporal co-activation patterns that are critical for texture-dominant effects (e.g., tremolo modulation or signal-dependent distortion).

To bridge these gaps, we study a retrieval-grounded preset-selection setting and focus on its core retrieval component: **Texture Resonance Retrieval (TRR)**. TRR models audio

style using Gram matrices of deep feature activations, explicitly capturing the *second-order correlations* that define temporal texture. We use this representation as a texture-aware retrieval prior: its role is to bias search toward executable presets that preserve effect-relevant structure, not to claim a universal metric of timbral or parameter-space similarity. The empirical evaluation in this paper is limited to a held-out guitar-effect benchmark.

Concretely, this paper makes two main evidence-backed contributions and several explicitly bounded diagnostics. First, we formulate editable guitar-effect preset selection as a retrieval-grounded problem in which outputs remain executable and inspectable inside DAW workflows. Second, we introduce **Texture Resonance Retrieval (TRR)** as a texture-aware retrieval prior, showing that second-order statistics (Gram matrices) improve retrieval over selected audio embedding baselines on the audited P0 Protocol-A guitar-effect benchmark and that controlled mechanism ablations favor Gram-style aggregation over same-backbone mean pooling. We further report near-duplicate sensitivity, a supervised direct-regression boundary, perception-alignment diagnostics, and real-degradation fusion stress tests as scope checks. These diagnostics are not used to claim that broad audio effect control has been solved; they define the boundary under which TRR is currently supported.

II. RELATED WORK

Our work lies at the intersection of intelligent music production, neural audio synthesis, and retrieval-based parameter control. We review each area and position our contributions accordingly.

Intelligent Music Production.

The challenge of translating perceptual descriptions into technical parameters was formally characterized by Stables et al. [4]. Their SAFE project demonstrated the feasibility of semantic-to-parameter mapping in a DAW context, but did not resolve the generalization problem across instruments, effect chains, and stylistic sub-genres. Schedl et al. [5] provides a comprehensive survey of music information retrieval techniques, highlighting semantic auto-tagging as a key challenge.

Early work in interactive control focused on learning intuitive mappings for synthesizers. Huang et al. [6] proposed active learning with Gaussian processes to discover perceptual control knobs, while more recently, Brade et al. [7] introduced SynthScribe, a multimodal tool for synthesizer sound retrieval. While these works advance interface design, they primarily target synthesis parameters (e.g., oscillator shape) rather than the complex signal chains of audio effects.

Audio Analysis Approaches. Subsequent work pursued automation via audio content analysis and differentiable mixing systems [8]. While effective when a target mix or reference audio already exists, this direction implicitly assumes the target audio is given. It therefore struggles to support creative timbre design from a purely conceptual request (“make it more vintage breakup”) and often suffers from an **opacity problem**: users see outputs but lack interpretable, editable rationale.

Recent advances in differentiable DSP have enabled neural approaches to audio effect modeling. Peladeau and Peeters [9] proposes blind estimation of audio effect parameters using auto-encoder architectures, addressing the inverse problem of parameter recovery. Liu et al. [10] introduces DDSP-SFX, which leverages differentiable digital signal processing for acoustically-guided sound effects generation, combining neural networks with traditional DSP modules for controllable synthesis. De Man and Reiss [11] explores generative latent spaces for neural audio synthesis, bridging DSP-informed and data-driven approaches. In parallel, style transfer methods have been applied to audio effects, with Steinmetz et al. [12] introducing ST-ITO for controlling audio effects via inference-time optimization using pretrained audio representations.

Our Positioning. Unlike prior work that treats parameter prediction as a *regression problem* or a purely generative problem, we frame it as a *retrieval task* that locates relevant, previously validated presets from a knowledge base and preserves editability in downstream DAW workflows.

Neural Audio Synthesis.

Neural audio synthesis has advanced rapidly, enabling high-fidelity text-to-audio generation [1, 2, 13, 14]. Diffusion models have emerged as the dominant paradigm, with Huang et al. [15] introducing prompt-enhanced text-to-audio generation, Wang et al. [16] achieving efficient high-quality synthesis via latent consistency models, and Ghosal et al. [17] exploring instruction-guided latent diffusion for better controllability. However, these models primarily output *finalized waveforms*, which are difficult to edit within professional workflows.

Accordingly, we view parameter-level generation as a **complement to, not a replacement for**, waveform-level synthesis: waveform-level generation offers broad expressivity but limited DAW editability, whereas parameter-level control preserves transparency and iterative refinement at the cost of requiring executable effect-chain priors.

In parallel, differentiable DSP (DDSP) [3] demonstrates that learned models can control interpretable synthesis parameters, and self-supervised audio representation learning (e.g., Wav2Vec 2.0 [18]) provides strong acoustic embeddings. Our retrieval-grounded formulation leverages these representations for similarity grounding, but targets editable parameter outputs compatible with DAW-centric workflows.

Audio Representation Learning. Recent advances in audio representation learning have produced strong embeddings for semantic tasks. CLAP [19] achieves robust text-audio alignment through contrastive pretraining, while PaSST [20] and PANNs [21] provide powerful acoustic features for classification and tagging. While these representations excel at semantic understanding, they are primarily designed for classification and captioning tasks. Our work adapts audio representations

(Wav2Vec2) for a different purpose: capturing timbral texture for parameter retrieval, where second-order statistics (Gram matrices) complement the first-order semantic embeddings. We use mid-level Wav2Vec2 features rather than CLAP-style single-vector embeddings because frame-level activations enable Gram matrix computation for texture characterization.

Retrieval-Grounded Parameter Control.

Retrieval-augmented generation (RAG) [22] is useful shorthand for systems that ground outputs in an external store, but the present paper studies a narrower setting: retrieval-grounded parameter control over executable presets. Our emphasis is not on open-ended text generation, but on retrieving feasible parameter exemplars that can be inspected and edited by the user. This distinguishes our setting from waveform generators and from document-centric multimodal RAG systems.

a) *Perceptual similarity vs. parameter transferability.*

Even strong general-purpose audio representations such as CLAP- or PaSST-style embeddings face a fundamental challenge: perceptual similarity does not necessarily imply that the underlying effect parameters are transferable. Recent audio-language models have improved zero-shot classification and captioning [19, 23–25], but they primarily target discrete prediction tasks rather than continuous parameter estimation. Instruction-following audio language models further broaden audio-text understanding [26, 27], yet executable effect-chain control remains distinct because the system must respect plugin constraints and feasible continuous control ranges. Our work addresses this gap by using second-order statistics (Gram matrices) to capture texture-relevant features that better align with parameter-space proximity.

Audio Texture and Semantic Audio Retrieval.

The concept of audio texture has been studied extensively in the context of style transfer and synthesis. Gatys et al. [28] introduced Gram matrix-based style representations for images, which were later adapted to audio by Bruna and Mallat [29] and others. Recent surveys on audio texture analysis [30] and audio representations for sound synthesis [31] highlight the importance of capturing second-order statistical features for timbre characterization. In contrast to prior uses primarily for texture synthesis or style transfer, we use Gram statistics as a retrieval embedding to locate executable parameter exemplars, treating retrieval as a prior for downstream constrained reasoning.

In semantic audio retrieval, the WikiMuTe dataset [32] provides a rich resource for connecting natural language descriptions with music audio, while retrieval-guided approaches [33] demonstrate the value of combining retrieval with generation for music captioning. Retrieval-augmented text-to-music generation [34] further validates the effectiveness of RAG approaches in audio tasks.

Our Texture Resonance Retrieval (TRR) builds upon these insights by using Gram matrices to capture audio texture through second-order statistics, thereby providing a more texture-sensitive retrieval prior for parameter inference on the current benchmark. The novelty claimed here is therefore not the Gram matrix itself, which is an established texture statistic, but its task-specific use as an executable preset-retrieval prior whose value is tested against audio embedding baselines,

leakage audits, and mechanism ablations in a parameter-control benchmark.

Table I summarizes our positioning. Our distinctive contribution is *texture-aware* audio retrieval via TRR for executable parameter control.

III. PROBLEM FORMULATION

We formalize audio effect parameter retrieval as a constrained synthesis problem: given a user query and an input signal, produce a parameter vector that is (i) *semantically consistent* with the user’s intent and (ii) *executable* by a real-time DSP engine. Unlike waveform generation, parameter synthesis must respect strict plugin-defined validity constraints, and the inverse mapping from perceptual language (e.g., “warm”, “punchy”) to a high-dimensional control vector is inherently ill-posed. Our goal is therefore not to claim a globally optimal solution in Θ , but to produce a feasible point estimate that is well-defined, interpretable, and grounded by concrete executable prototypes. For empirical evaluation, we instantiate Φ and Θ with a multi-effect guitar processing chain.

A. Signal Processing Dynamics

Consider a (potentially non-differentiable) DSP rendering operator $\Phi : \mathcal{X} \times \Theta \rightarrow \mathcal{X}$, where $\mathcal{X}$ denotes the space of discrete-time audio signals and $\Theta \subset \mathbb{R}^d$ denotes the d -dimensional control parameter space. Given an input signal $\mathbf{x}_{\text{in}} \in \mathcal{X}$, the operator produces an output $\mathbf{x}_{\text{out}} = \Phi(\mathbf{x}_{\text{in}}; \theta)$. In practical DAW and plugin systems, Φ can include nonlinearities, conditional branches (e.g., bypass logic), and internal state (e.g., smoothing), making end-to-end differentiation unreliable or inapplicable for direct optimization over θ .

The parameter space Θ is not a free vector space but a bounded hyper-rectangle subject to structural constraints:

$$\Theta = \{\theta \in \mathbb{R}^d \mid \mathbf{l} \preceq \theta \preceq \mathbf{u}, \quad \mathcal{C}(\theta) = 1\} \quad (1)$$

where $\preceq$ denotes element-wise inequality, $\mathbf{l}, \mathbf{u}$ define physical operating ranges (e.g., frequency bounds [20, 20k]), and $\mathcal{C} : \mathbb{R}^d \rightarrow \{0, 1\}$ represents a binary validity function encoding inter-parameter dependencies (e.g., ensuring filter resonance $Q > 0$). In practice, DSP parameter spaces can be non-convex and partially discrete (e.g., on/off switches or mode selectors), so we use Eq. 1 as a pragmatic executable abstraction rather than a claim of convexity or smoothness. This motivates our reliance on retrieval-grounded synthesis and deterministic validity enforcement instead of direct gradient-based optimization over Θ .

B. Goal and Constraints

The user’s intent is encapsulated in a multi-modal query $q = (t, \mathbf{a}_{\text{ref}})$, comprising a natural language description $t \in \mathcal{T}$ and an optional audio reference $\mathbf{a}_{\text{ref}} \in \mathcal{X} \cup \{\emptyset\}$. Our objective is to produce a *valid* parameter configuration that is consistent with the query:

$$\tilde{\theta} \in \Theta \quad \text{given } q, \quad (2)$$

where validity is defined by the plugin-specific constraint function $\mathcal{C}(\theta) = 1$ and physical bounds $\mathbf{l} \preceq \theta \preceq \mathbf{u}$ (Eq. 1). We

use θ^* to denote an ideal (possibly non-unique) solution that best matches the user’s perceptual intent; in practice we only have access to retrieval-based approximation, and therefore aim for a feasible point estimate $\tilde{\theta}$.

a) *Operational objective (without probabilistic semantics).*: The query-consistency of $\tilde{\theta}$ is not directly measurable as a differentiable loss, because the target is perceptual and depends on a complex signal chain. Accordingly, we adopt a retrieval-grounded operational objective: construct an executable neighborhood in Θ using a finite preset database, and then adapt within that neighborhood while enforcing feasibility. This yields a system whose intermediate artifacts (retrieved prototypes, draft parameters, constraint repairs) are auditable by design, which is critical for a journal setting emphasizing system rigor and reproducibility.

C. The Retrieval-Based Approach

Because the inverse mapping from perceptual intent to parameters is ill-posed and the DSP validity constraints are strict, we implement a retrieval-based pipeline. Let z denote a retrieved preset (a discrete prototype) from a finite database. We retrieve top- K candidates conditioned on q :

$$\theta_{\text{selected}} = \text{Top-}K(q) \quad (3)$$

Intuitively, retrieval supplies a coarse *mode-seeking* step that places the query in a plausible neighborhood of Θ .

a) *Design Principle 1 (Well-defined synthesis under missing modalities).*: Because $\mathbf{a}_{\text{ref}}$ is optional, the synthesis pipeline must remain well-defined when $\mathbf{a}_{\text{ref}} = \emptyset$. Our formulation treats q as a tuple with an explicit null element and requires every downstream stage (retrieval and projection) to specify a deterministic fallback behavior when an input modality is absent.

IV. METHODOLOGY

We implement the retrieval-based framework defined in Section III via **Texture Resonance Retrieval (TRR)** to retrieve candidate presets using second-order texture statistics.

A. System Architecture

The system realizes the operator $\Phi(\mathbf{x}_{\text{in}}; \theta)$ through a decoupled design, visualized in Figure 1.

The architecture consists of two main components. (i) **The real-time DSP engine** implements Φ as a low-latency audio processing graph and executes parameters retrieved from the knowledge base. We denote the parameter state consumed by the DSP engine as θ_{safe} , which can be smoothed over time to avoid zipper noise and audible artifacts. (ii) **The retrieval module** implements the mapping $q \rightarrow \theta_{\text{safe}}$ by performing dual-modal search (text and audio) asynchronously at a lower rate than the audio callback, returning validated parameters that are guaranteed to satisfy plugin constraints.

a) *Design Principle 2 (Runtime isolation).*: At inference time, the real-time audio path is a deterministic function of $(\mathbf{x}_{\text{in}}, \theta_{\text{safe}})$ through $\Phi(\mathbf{x}_{\text{in}}; \theta_{\text{safe}})$. The inference agent does not execute inside the real-time callback and can only influence audio through asynchronous updates of θ_{safe} after validity checking.

TABLE I
POSITIONING AGAINST PRIOR WORK

Work	Input	Output	Grounding	Editability
SAFE [4]	Text	Parameters	None	Yes
DDSP [3]	Audio	Parameters	None	Yes
LLM2Fx [35]	Text	Parameters	None	Yes
AudioLDM [1]	Text	Waveform	Pretrained	No
HM-RAG [36]	Text+Doc	Text	Retrieval	N/A
TRR-based retrieval control (Ours)	Text+Audio	Parameters	Dual-Modal Retrieval	Yes

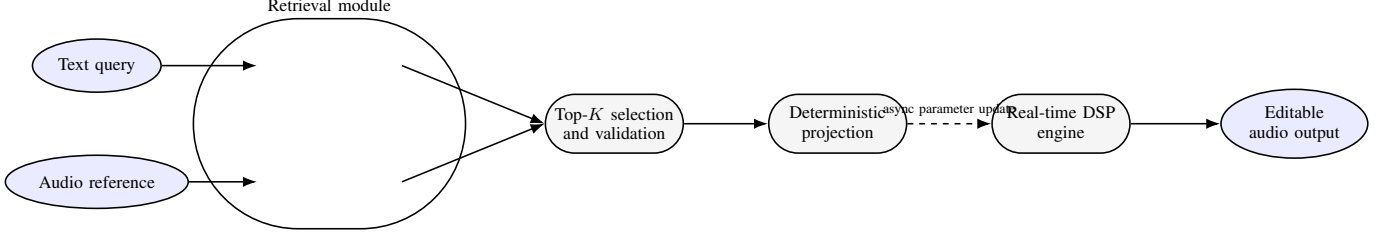


Fig. 1. System overview of the retrieval-grounded guitar-effect preset-selection pipeline over a switchable nine-module topology: Compressor → Driver → Screamer → Equaliser → Chorus → Flanger → Phaser → Delay → Reverb.

B. Texture Resonance Retrieval (TRR)

The core challenge in texture-aware retrieval lies in computing a similarity metric for the audio modality that preserves effect-relevant dynamics. Standard embeddings (e.g., mean-pooled Wav2Vec2 features) collapse temporal structure into a static vector, losing information that is critical for effects like modulation and distortion.

To address this, we introduce **Texture Resonance Retrieval (TRR)**. Drawing on the hypothesis that audio texture is encoded in second-order feature statistics [37], we compute the Gram matrix of deep activations.

Algorithm 1 TRR Encoding

Require: Audio sample a , Wav2Vec2 model $\mathcal{W}$, layer set $\mathcal{L} = \{4, 5, 6\}$, frozen random linear projection $P : \mathbb{R}^{768} \rightarrow \mathbb{R}^{32}$

Ensure: Texture embedding $\mathbf{g}_{\text{norm}}$

1. Preprocessing: Normalize a and pad to minimum length.

for each $\ell \in \mathcal{L}$ **do**

$\mathbf{H}_\ell \leftarrow \mathcal{W}_{\text{layer } \ell}(a) \in \mathbb{R}^{T_{\text{frm}} \times 768}$

$\hat{\mathbf{H}}_\ell \leftarrow P(\mathbf{H}_\ell) \in \mathbb{R}^{T_{\text{frm}} \times 32}$

$\mathbf{G}_\ell \leftarrow \frac{1}{T_{\text{frm}}} \hat{\mathbf{H}}_\ell^\top \hat{\mathbf{H}}_\ell \in \mathbb{R}^{32 \times 32}$

end for

$\bar{\mathbf{G}} \leftarrow \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{G}_\ell$

$\mathbf{g}_{\text{norm}} \leftarrow \text{vec}(\bar{\mathbf{G}}) / \|\text{vec}(\bar{\mathbf{G}})\|_2$

return $\mathbf{g}_{\text{norm}}$

Let $\mathbf{H}_\ell \in \mathbb{R}^{T_{\text{frm}} \times 768}$ be the frame-level activation map from layer ℓ of a pre-trained Wav2Vec2 Base model. In the benchmarked implementation we use the mid-level layer set $\mathcal{L} = \{4, 5, 6\}$, apply a frozen Xavier-initialized random projection to 32 dimensions, compute a Gram matrix per layer, average the resulting 32×32 matrices, flatten the result into a

1024-dimensional vector, and L2-normalize it. The averaged texture descriptor can be written as

$$\bar{\mathbf{G}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{1}{T_{\text{frm}}} \hat{\mathbf{H}}_\ell^\top \hat{\mathbf{H}}_\ell, \quad (4)$$

where $\hat{\mathbf{H}}_\ell$ denotes the projected feature map. The entry $\mathbf{G}_{ij}$ captures the co-activation of projected channels i and j irrespective of their absolute temporal position. This time-invariance allows TRR to match textures (e.g., “fast tremolo”) even if the reference and database entries are not phase-aligned. L2 normalization of $\text{vec}(\bar{\mathbf{G}})$ is equivalent to Frobenius normalization of the matrix representation, and similarity is then computed as cosine similarity between the flattened descriptors. Computationally, Gram-matrix construction scales as $O(D^2 \cdot T_{\text{frm}})$ after projection, where $D = 32$ in the reported implementation. This bound covers only the post-Wav2Vec2 Gram construction; the Wav2Vec2 forward pass is a separate feature-extraction cost and dominates when features are not cached.

a) Properties and limitations.: TRR can be viewed as a second-order summary of deep features: it retains cross-channel correlation structure while discarding absolute temporal alignment. This is desirable for retrieval in effect spaces where perceptual “style” is often expressed through repeated modulation patterns or stationary texture rather than a single time-locked event. The Frobenius normalization further reduces sensitivity to global scale changes in feature activations, making cosine similarity comparisons less dominated by magnitude. However, TRR is not a universal representation: by aggregating over time, it intentionally discards fine-grained ordering and can under-represent transient-dominant effects. Moreover, as with other deep representations, retrieval behavior can depend on the chosen backbone layer and the upstream feature extractor.

TRR Gram Matrix Heatmaps (Representative Queries)

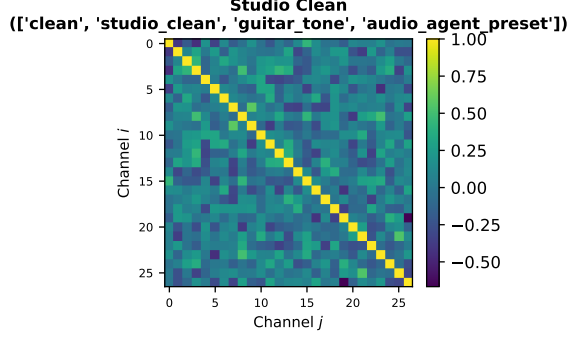


Fig. 2. Representative TRR Gram-matrix examples illustrating the second-order co-activation structure used by the retrieval descriptor.

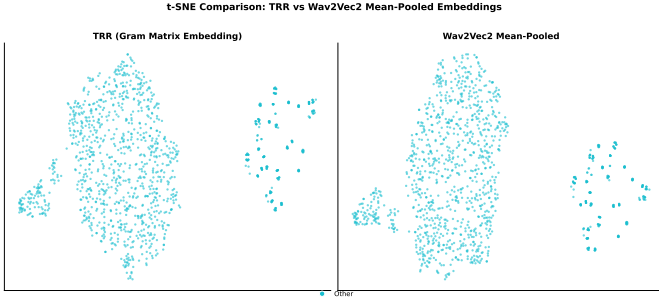


Fig. 3. t-SNE visualization comparing TRR and mean-pooled Wav2Vec2 embeddings on the benchmark presets. The plot is used only as a qualitative representation diagnostic.

The alignment-insensitivity statement is a direct consequence of Gram aggregation: for a frame permutation matrix $\mathbf{P}$, $\frac{1}{T_{\text{frm}}} \mathbf{H} \mathbf{H}^T = \frac{1}{T_{\text{frm}}} (\mathbf{H} \mathbf{P}) (\mathbf{H} \mathbf{P})^T$. We use this as an implementation motivation rather than as an independent theoretical contribution.

Figure 2 visualizes representative TRR Gram matrices, and Figure 3 compares the resulting TRR embedding geometry with mean-pooled Wav2Vec2 features. These figures are diagnostic visualizations of the representation, not separate quantitative evidence.

C. Complexity and Latency Considerations

Journal readers often care not only about conceptual correctness but also about operational constraints. Our design separates (a) *real-time execution*, which must remain deterministic and low-latency, from (b) *asynchronous inference*, which can be slower and is not executed on the audio callback. On the inference side, TRR encoding includes Wav2Vec2 feature extraction, then $O(D^2 T_{\text{frm}})$ Gram construction after projection with $D = 32$ (Section IV-B), followed by standard top- K nearest-neighbor search. The retrieval system operates asynchronously from the real-time audio path, ensuring that audio processing latency is not affected by retrieval operations.

a) *Latency measurement protocol.*: We report a reproducible *offline* latency breakdown for the local retrieval path

TABLE II
PER-QUERY CACHED RETRIEVAL-SCORING LATENCY BREAKDOWN. WE REPORT MEDIAN AND P95 OVER 633 RUNS FROM THE ORIGINAL 211-QUERY PROFILING POOL (211×3 REPEATS). THESE NUMBERS EXCLUDE FEATURE EXTRACTION, LIVE LLM CALLS, PLUGIN RENDERING, AND USER-INTERFACE OVERHEAD, AND SHOULD NOT BE INTERPRETED AS END-TO-END DAW REAL-TIME LATENCY.

Component	Median (ms)	p95 (ms)
Text scoring	3.027	3.586
Audio scoring (TRR cosine)	1.369	1.564
Fusion	0.110	0.146
Projection	0.031	0.039
End-to-end (local pipeline)	4.538	5.269

under a cache-only profiling setup. Measurements were collected on an Apple Silicon machine (arm64 CPU, 10 cores; macOS; Python 3.12) over the original 211-query profiling pool, with 5 warm-up iterations and 3 repeats per query. We report median and p95 over the resulting 633 timed query runs. These measurements exclude live network LLM latency and should therefore be interpreted as local retrieval-path timing rather than an end-to-end cloud deployment budget.

The retrieval system operates asynchronously from the real-time audio callback, ensuring that audio processing latency is not affected by retrieval operations.

D. Multimodal Fusion as a Supplementary Diagnostic Extension

Beyond the core retrieval benchmark, the repository contains an experimental fusion mechanism that combines text-side and audio-side scores under degraded-input conditions. In the current codebase this mechanism is implemented as a practical quality-aware fallback rule rather than a single clean probabilistic model, so we treat its results as a supplementary diagnostic stress test rather than as one of the paper’s main validated contributions. The central evidence chain of the manuscript therefore remains the retrieval-only benchmark in Protocol-A. For candidate preset z_i , the fixed-weight diagnostic uses

$$s_{\text{fused}}(z_i) = \alpha s_{\text{text}}(z_i) + (1 - \alpha) s_{\text{audio}}(z_i), \quad (5)$$

where α denotes the text-side weight after score normalization. The adaptive profile changes α according to degradation metadata and is evaluated only as a robustness diagnostic.

V. EXPERIMENTS

We validate the proposed framework by answering one core research question: does modeling second-order texture statistics (TRR) improve parameter inference compared to first-order baselines?

A. Experimental Setup

a) *Dataset and Knowledge Base.*: Our full server-side corpus contains 1267 paired records (audio–text–parameter triplets) collected from guitar effect presets. A post-hoc audit of the original song-name-based held-out list (211 test queries, 1056 knowledge-base items) revealed 16 cross-split shared

resolved audio paths and 48 near-duplicate cross-split pairs under a cheap fingerprint. To remove exact audio reuse from the main objective benchmark, we regroup records by resolved local audio path before sampling the held-out set. The resulting stricter Protocol-A split contains 204 test queries and 1063 knowledge-base items, with zero exact shared resolved audio paths across split under the current audit. We retain this as a pilot benchmark because cheap near-duplicate pairs remain and the corpus is still guitar-only.

b) Provenance notes.: The dataset JSON exposes song name, style tags, feature tags, parameters, and stored vectors, but it does not include a separate auditable text-source field. We therefore treat the style and feature fields as the only explicit text-side provenance available in-repo, and use the stored parameters as the operational ground truth.

c) Scope and protocol.: Unless otherwise stated, objective results in this section use **P0 Protocol-A** (retrieval-only) on the shared 204-query split and use top-1 retrieval for all retrieval methods. Protocol-C, listening tests, and latency measurements are reported on their own protocol-specific pools and are labeled explicitly.

d) Resources and reproducibility.: To facilitate reproducibility, we provide an anonymized supplementary package for review containing preprocessing details, embedding computation steps, retrieval-index construction, prompt templates, evaluation scripts, and split specifications for all reported tables. The cleaned repository and artifacts will be released publicly upon acceptance.

e) Baselines.: We compare our proposed **TRR** (Section IV-B) against five retrieval baselines in the P0 benchmark. **Wav2Vec** uses mean-pooled Wav2Vec2 activations. **FeatureNN** uses the released benchmark feature vectors and should therefore be interpreted as a fixed dense-retrieval reference. **CLAP**, **PaSST**, and **PANNs** use pretrained audio representations as stronger external embedding baselines for retrieval. Earlier Text-RAG and resolved-audio-grouped diagnostic rows are retained only in the supplementary audit trail and are not the main P0 baseline matrix.

B. Metrics

We evaluate alignment in parameter space Θ using operational metrics computed on flattened numeric parameter leaves. Concretely, given a nested parameter dictionary, we recursively flatten numeric leaves into a sparse vector indexed by dot-separated keys (e.g., `DelayOn.Mix`) and ignore non-numeric leaves (e.g., strings). Unless otherwise stated, we form the union of keys between prediction and ground truth, and treat missing numeric keys as zeros for metric computation. The main P0 table reports raw L2, normalized L2, Acc@0.1, Recall, Cosine, and Module consistency; the paired significance tests use per-query normalized L2. Raw L2 magnitudes can be dominated by a small number of large-scale parameters and should be interpreted within each protocol rather than across protocols. Normalized L2 uses plugin-defined min-max ranges for comparability, but this is still an engineering normalization rather than a perceptual scale: logarithmic, dB, and nonlinear control laws can make equal normalized parameter differences perceptually unequal.

TABLE III
P0 PROTOCOL-A: DIRECT RETRIEVAL COMPARISON ON THE SHARED 204-QUERY BENCHMARK. LOWER IS BETTER FOR L2 AND NORM.L2; HIGHER IS BETTER FOR ALL OTHER METRICS. TRR ACHIEVES THE BEST MEAN PERFORMANCE ON THE NORMALIZED DISTANCE AND ALL REPORTED ALIGNMENT SCORES.

Method	n	L2 ↓	Norm.L2 ↓	Acc@0.1 ↑	Recall ↑	Cos ↑	Module ↑
TRR (Ours)	204	8.7287	0.1881	0.5766	0.5404	0.8415	0.8297
Wav2Vec	204	22.5174	0.2400	0.5015	0.4648	0.5642	0.7423
FeatureNN	204	17.9391	0.2236	0.5264	0.4779	0.6030	0.7544
CLAP	204	16.1257	0.2298	0.5389	0.4922	0.6531	0.7837
PaSST	204	18.2860	0.2132	0.5422	0.5001	0.6192	0.7960
PANNs	204	21.0269	0.2260	0.5090	0.4759	0.5826	0.7606

a) L2 Error (RMSE).: Given flattened vectors $v(\theta_{\text{gt}})$ and $v(\tilde{\theta})$ over the union key set of size d , we report the root mean squared error

$$\text{L2} = \sqrt{\frac{1}{d} \sum_{i=1}^d \left(v(\theta_{\text{gt}})^{(i)} - v(\tilde{\theta})^{(i)} \right)^2}. \quad (6)$$

b) Accuracy@0.1.: We report the fraction of flattened numeric parameters whose absolute error is within a fixed tolerance $\tau = 0.1$:

$$\text{Acc@0.1} = \frac{1}{d} \sum_{i=1}^d \mathbb{I} \left(\left| v(\theta_{\text{gt}})^{(i)} - v(\tilde{\theta})^{(i)} \right| \leq 0.1 \right). \quad (7)$$

c) Recall (active numeric parameters).: We define a ground-truth numeric parameter as *active* if $|v(\theta_{\text{gt}})^{(i)}| > 0.05$. Recall is the fraction of active ground-truth dimensions whose predicted value falls within the same tolerance $\tau = 0.1$.

d) Cosine Similarity.: We compute cosine similarity between flattened vectors over the union key set:

$$\text{Cosine} = \frac{v(\theta_{\text{gt}}) \cdot v(\tilde{\theta})}{\|v(\theta_{\text{gt}})\|_2 \|v(\tilde{\theta})\|_2}. \quad (8)$$

e) Module Consistency.: We compute module-level consistency as Jaccard similarity over active module sets inferred from keys ending with `On`:

$$\text{Module} = \frac{|M_{\text{gt}} \cap M_{\text{pred}}|}{|M_{\text{gt}} \cup M_{\text{pred}}|}. \quad (9)$$

C. Protocol-A: Comparative Retrieval Evaluation

Purpose. Benchmark TRR against the P0 retrieval baselines for executable parameter control under the shared 204-query Protocol-A split.

Protocol. All results in this subsection follow P0 Protocol-A (retrieval-only) on a deterministic 204-query test split and 1,063-record retrieval database. All methods use top-1 retrieval. In addition to Wav2Vec, FeatureNN, and CLAP, this P0 evidence package adds PaSST and PANNs as stronger acoustic baselines. We report both raw L2 and normalized L2, but use normalized L2 for the paired significance tests because the P0 statistical report defines the test on per-query normalized parameter distance.

Table III reports the main comparison across the current retrieval baselines.

TABLE IV

NEAR-DUPLICATE SENSITIVITY FOR TRR ON THE AUDITED PROTOCOL-A SPLIT. THE THRESHOLD CONTROLS CHEAP-FINGERPRINT FILTERING BETWEEN HELD-OUT QUERIES AND THE RETRIEVAL DATABASE. LOWER NORM.L2 IS BETTER.

Threshold	Removed	KB size	Norm.L2 ↓	Cos ↑
0.000	0	1063	0.1879	0.8149
0.005	594	469	0.2026	0.8169
0.010	654	409	0.1993	0.8442
0.020	903	160	0.2265	0.7758
0.050	980	83	0.2493	0.7024

Result. On the P0 split, TRR is best on Norm.L2, Acc@0.1, Recall, Cosine, and Module in Table III. The strongest non-TRR rows are CLAP and PaSST, but neither surpasses TRR on the principal normalized and alignment metrics. Paired Wilcoxon tests on per-query Norm.L2 reject the TRR-vs-baseline null hypothesis after Holm correction for Wav2Vec ($p = 1.564 \times 10^{-7}$), FeatureNN ($p = 9.523 \times 10^{-5}$), CLAP ($p = 4.476 \times 10^{-4}$), PaSST ($p = 1.254 \times 10^{-3}$), and PANNs ($p = 2.122 \times 10^{-5}$). The corresponding effect sizes are small, so we interpret the result as statistically reliable but moderate in magnitude rather than as a universal dominance claim.

The practical interpretation is deliberately narrow. TRR is the strongest *currently evaluated* retrieval prior on this benchmark because its second-order texture statistics better align with executable parameter neighborhoods than first-order or classification-oriented embeddings. The small effect sizes in the normalized-distance tests also matter: they indicate that the gain is not a uniform re-ordering of every query, but a consistent shift toward better parameter alignment across paired examples. We therefore treat TRR as a stronger retrieval prior for texture-sensitive control, not as a universal best-method claim beyond the present protocol.

a) Near-duplicate sensitivity. Because retrieval benchmarks are especially sensitive to near-duplicate leakage, we additionally stress-test the result by filtering cheap-fingerprint near-duplicate pairs between the held-out queries and the retrieval database. Table IV reports the TRR row as progressively stricter thresholds remove more candidate items from the database. We report only normalized and directional metrics in this table because the sensitivity script recomputes candidate filtering with a script-local raw parameter pass; retaining raw L2 there would invite an invalid comparison with the main Protocol-A table. TRR remains stronger than the evaluated non-TRR retrieval baselines in the corresponding sensitivity report, but its normalized L2 degrades as the candidate pool becomes very small. We therefore interpret the main result as robust to the removal of many cheap near duplicates, but not as evidence that the current benchmark is free of all duplicate-family risk.

b) Direct-regression boundary. To test whether retrieval is merely hiding an easier supervised parameter-estimation problem, we also run an MLP regressor trained on the P0 knowledge-base items using mean-pooled Wav2Vec2 vectors as inputs, followed by a minimal validity projection that clips

TABLE V

SUPERVISED RANGE-PROJECTED REGRESSION BOUNDARY ON THE P0 SPLIT. THE MLP IS STRONGER ON NORMALIZED NUMERIC ERROR AND ACC@0.1, WHILE TRR PRESERVES RETRIEVED-EXEMPLAR SEMANTICS AND IS STRONGER ON RECALL AND COSINE.

Method	n	L2 ↓	Norm.L2 ↓	Acc@0.1 ↑	Recall ↑	Cos ↑
TRR retrieval	204	8.7287	0.1881	0.5766	0.5404	0.8415
MLP + range projection	204	57.6239	0.1603	0.7894	0.3468	0.5941

TABLE VI

P0 PROTOCOL-C: REAL-DEGRADATION FUSION ROBUSTNESS ON THE SHARED 204-QUERY SPLIT. LOWER IS BETTER FOR L2; HIGHER IS BETTER FOR ALL OTHER METRICS.

Method	α	L2 ↓	Acc@0.1 ↑	MRR ↑	NDCG@5 ↑
fixed fusion 0.30	0.3000	29.2322	0.4846	0.4095	0.3723
fixed fusion 0.50	0.5000	20.5551	0.6085	0.5607	0.5528
fixed fusion 0.70	0.7000	20.1868	0.6104	0.5652	0.5551
adaptive fusion	0.5980	20.1491	0.6121	0.5654	0.5566

each numeric parameter to the training-set observed range. The model is a two-layer MLP with 256 hidden units trained on 1,058 usable records and evaluated on the same 204 held-out queries. As shown in Table V, this projected supervised boundary improves normalized L2 and Acc@0.1, but has substantially weaker cosine and recall than TRR and does not return a retrieved exemplar. Thus, if the sole objective is normalized numeric parameter error, the simple supervised regressor is the stronger boundary. Our claim is correspondingly narrower: TRR provides retrieved exemplars with a moderate parameter-space trade-off and better neighborhood semantics than the evaluated retrieval baselines, not a dominance result over direct regression.

D. Diagnostic: Degraded-Input Fusion

Purpose. Evaluate whether multimodal retrieval remains stable when the audio reference is realistically degraded before representation extraction.

Protocol. P0 Protocol-C reuses the same 204-query split as Protocol-A. The audio branch is degraded in waveform space and then re-encoded into TRR embeddings. The degradation set includes AWGN at 20/10/5 dB, MP3 at 128/64/32 kbps, reverb with RT60 0.6/1.0, and truncation to 50%/30% of the signal. The text branch uses Style+Feature, and ranking metrics use an oracle relevance set defined by the nearest 10% of knowledge-base items in parameter space, with a minimum of five relevant items per query. We compare fixed fusion weights against an adaptive fusion profile; the pure-text boundary is retained only as an auxiliary diagnostic and is not part of the main multimodal comparison.

Diagnostic result. Table VI shows that audio-heavy fusion is the least stable operating point under realistic degradations: fixed fusion with $\alpha = 0.30$ yields L2=29.2322 and the weakest ranking scores. Mid-to-high text-weight fusion is substantially more robust. Adaptive fusion obtains the best aggregate L2, Acc@0.1, MRR, and NDCG@5 among the reported multimodal settings, but the margin over fixed $\alpha = 0.50$ and $\alpha = 0.70$ is small. We therefore move Protocol-C outside the

TABLE VII
TRIAL 1 (STYLE MATCHING): CONDITION-LEVEL STATISTICS
(WITHIN-SUBJECT MEANS).

Condition	Mean	Median	Std
reference	86.85	92.50	16.70
Jazz Clean	76.19	81.00	20.80
Phase	75.38	80.50	20.88
Blues Solo	72.96	69.50	21.19
Ambient Guitar	72.35	73.50	18.45
Flanger	69.12	74.00	23.08
Modern Metal	65.58	70.00	30.22
Chorus	64.92	66.50	23.12

main evidence chain: it is useful for showing that degraded-input retrieval is a modality-weighting problem, but it does not establish adaptive fusion as a generally superior multimodal method.

E. Perceptual Listening Test (Multiple-Stimulus with Hidden Reference)

Protocol. We conduct a multiple-stimulus listening test with a hidden reference with 26 participants (18 female, 8 male; self-reported ages available for 25 participants with mean 22.2 years and range 18–30), yielding 910 ratings across 10 trials. Each trial uses a continuous 0–100 slider (higher is better), presents stimuli in randomized order, and includes the ground-truth preset rendering as a hidden reference (`reference`); we do not include an explicit low-quality anchor required by ITU-R BS.1534, and therefore avoid referring to this study as a standard MUSHRA test. Trial 1 evaluates style matching across seven labeled target conditions, Trials 2–5 compare our TRR-based system against manual parameter tuning, and Trials 6–10 compare the TRR-based system against a waveform-generation baseline (MusicGen). Statistical analysis uses repeated-measures nonparametric tests (Friedman omnibus; post-hoc Wilcoxon signed-rank with Holm correction) on raw participant ratings. The repository logs preserve participant IDs and scores, but they do not record device metadata, musical expertise, or manual-tuning provenance/time budgets; accordingly, the listening study complements the retrieval benchmark but should not be interpreted as a fairness-critical workflow trial or as a direct confirmation of every system extension discussed elsewhere in the manuscript.

1) *Trial 1: Style Matching.* **Data.** 208 ratings (26 participants $\times$ 8 conditions). Overall mean 72.92 (median 77.50, std 22.72).

2) *Trials 2–5: Guitar Solo Comparison.* **Data.** 312 ratings (26 participants $\times$ 3 conditions $\times$ 4 trials). Overall mean 69.53 (median 73.50, std 23.53). Within-subject means are 71.55 (TRR-based system) vs 51.72 (manual). The repeated-measures Friedman test is significant ($Q = 41.6154$, permutation $p = 5 \times 10^{-5}$), and the post-hoc TRR-vs-manual comparison remains significant after Holm correction ($p_{\text{Holm}} = 1.5 \times 10^{-4}$). However, because the logs do not preserve the manual-baseline operator, expertise, or time budget, this block should be read as supportive perceptual context rather than as a fairness-critical optimization comparison.

TABLE VIII
TRIALS 2–5 (SOLO COMPARISON): CONDITION-LEVEL STATISTICS
(WITHIN-SUBJECT MEANS).

Condition	Mean	Median	Std
reference	85.33	85.25	12.52
TRR-based system	71.55	72.88	12.41
manual	51.72	55.00	17.15

TABLE IX
TRIALS 6–10 (SIMILARITY): CONDITION-LEVEL STATISTICS
(WITHIN-SUBJECT MEANS).

Condition	Mean	Median	Std
reference	89.02	91.50	10.04
TRR-based system	42.31	41.90	19.30
MusicGen	41.29	43.90	19.58

3) *Trials 6–10: Similarity to Reference.* **Data.** 390 ratings (26 participants $\times$ 3 conditions $\times$ 5 trials). Overall mean 57.54 (median 59.00, std 32.24). The TRR-based system and MusicGen achieve comparable similarity scores: within-subject means are 42.31 (TRR-based system) vs 41.29 (MusicGen), while the reference condition scores 89.02. The Friedman omnibus remains significant ($Q = 36.2308$, permutation $p = 5 \times 10^{-5}$), but the TRR-vs-MusicGen pairwise comparison is not significant after Holm correction ($p_{\text{Holm}} = 0.7797$), so we interpret this block as evidence of parity rather than superiority.

a) *Interpretation.* These listening-test results are exploratory perceptual context and should not be read as validation of any single objective metric. Trial 1 shows that the best non-reference condition remains below the hidden reference, Trials 2–5 are limited by incomplete manual-baseline provenance, and Trials 6–10 show no significant TRR advantage over MusicGen on this similarity task. The listening study therefore motivates future query-aligned perceptual evaluation rather than proving perceptual superiority of the retrieval system.

b) *Metric-perception alignment diagnostic.* We additionally ran a conservative alignment diagnostic linking Trial-1 style labels to Protocol-A TRR rows through an explicit alias table and querying a local qwen3.5-9b vLLM server for text-side perceptual-recognizability scores. This analysis is intentionally weak: only four of the seven Trial-1 non-reference labels matched Protocol-A query names, and the local model scores are text-label judgments rather than audio judgments. Under this restricted setting, human mean ratings did not significantly correlate with TRR objective metrics: for human rating versus negative TRR L2, Spearman $\rho = 0.40$ with $n = 4$ and $p = 0.60$; for human rating versus TRR Acc@0.1, $\rho = 0.00$ with $n = 4$ and $p = 1.00$; for human rating versus TRR cosine, $\rho = 0.00$ with $n = 4$ and $p = 1.00$. The local vLLM score also failed to provide a reliable positive bridge to human ratings (human-vs-qwen3.5-9b score: Spearman $\rho = -0.18$, $n = 7$, $p = 0.7005$). We therefore use this diagnostic as a negative control: parameter-space alignment is the primary audited objective in this paper,

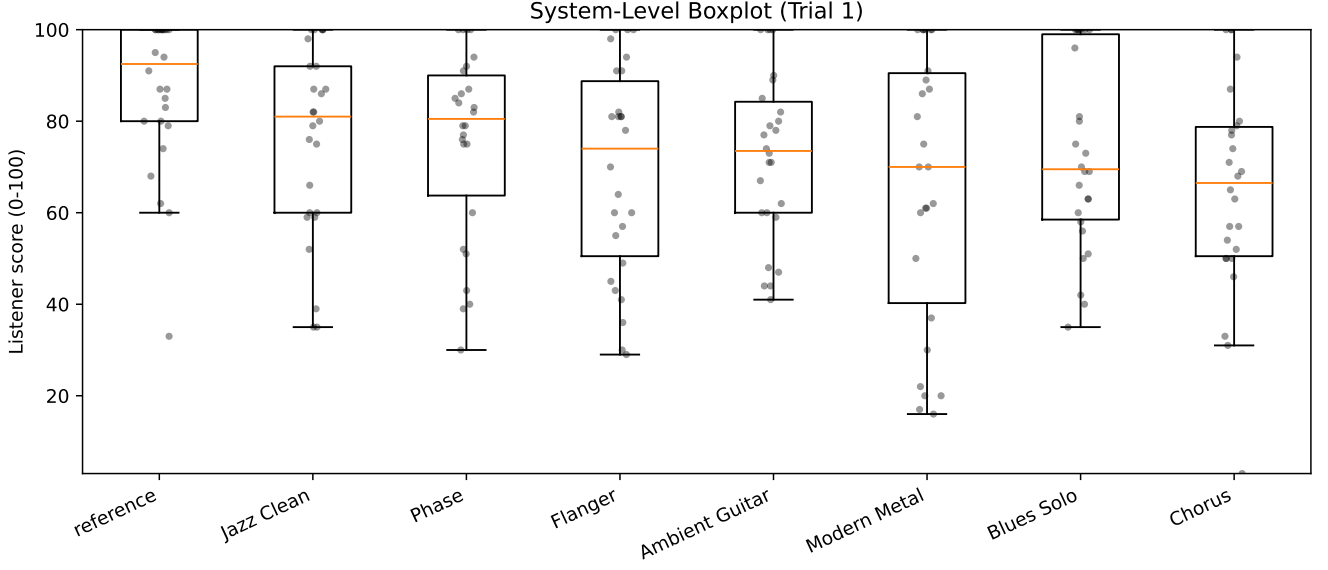


Fig. 4. Trial 1: condition-level score distributions (boxplot) for style matching.

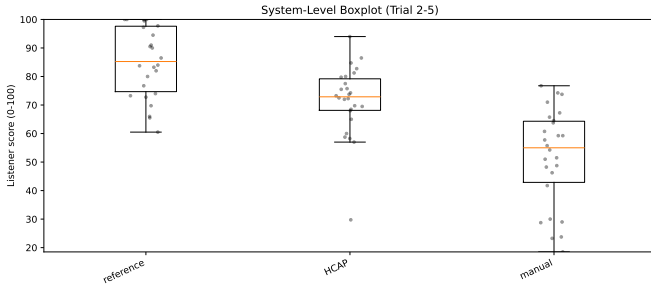


Fig. 5. Trials 2–5: condition-level score distributions.

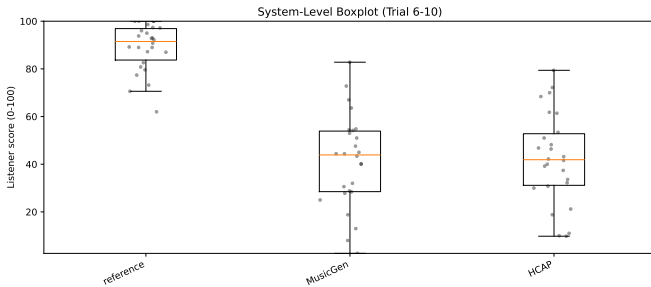


Fig. 6. Trials 6–10: condition-level score distributions.

while perceptual alignment remains unvalidated by the current evidence.

VI. DISCUSSION

A. Scope of the Current Evidence

The strongest evidence in the current manuscript is the P0 Protocol-A retrieval benchmark, supplemented by near-duplicate sensitivity, mechanism ablations, and the supervised

direct-regression boundary. Protocol-C and the listening/local-vLLM diagnostics provide useful stress tests, but they are not treated as primary evidence for broad audio-effect control. Personalization and memory components are outside the paper’s validated claims because they are not evaluated under the same held-out protocol.

B. Implications for Audio Effect Control

The current results suggest a practical pattern for retrieval-based audio control: use retrieval over real presets to provide feasibility constraints and stylistic priors, and then expose the retrieved configuration for direct user editing. In Protocol-A, TRR achieves the strongest parameter alignment in the reported retrieval comparisons while preserving editability and traceability in DAW workflows. The supervised MLP boundary with range projection is stronger on normalized L2 and Acc@0.1, so direct regression is a serious boundary rather than a weak baseline. Together, these results support TRR as a retrieved-exemplar prior with moderate parameter-space trade-offs, not a universally superior parameter estimator or a proof of full autonomous production.

C. Revisiting the Texture Hypothesis

In Protocol-A, TRR improves parameter-space alignment over the mean-pooled Wav2Vec baseline. This comparison controls for the acoustic backbone and changes the aggregation operator from first-order pooling to second-order Gram statistics. The supplementary mechanism ablation reinforces this interpretation: on the same P0 split, a same-layer Gram representation improves normalized L2 over mean pooling, the full unprojected Gram boundary also improves over mean pooling, and removing L2 normalization degrades retrieval quality. It therefore supports the hypothesis that correlation structure can be useful as a texture descriptor for retrieval.

The derived per-query analysis sharpens this claim: because many queries tie across methods, second-order structure should not be read as uniformly re-ordering the benchmark. Rather, it appears to matter most when coarse semantic similarity is insufficient and retrieval depends on texture-specific co-activation patterns. The ablation also shows that layer selection and projection dimension are not cosmetic details; the reported implementation is a practical operating point, not a proof of a globally optimal TRR architecture. We therefore interpret TRR as a *retrieval prior* that can be beneficial for texture-sensitive control, rather than a universal metric for parameter transfer.

An open direction is whether analogous second-order representations transfer to other audio-effect settings beyond the current benchmark, including additional instruments, effect families, and mix-stage tasks. Future work can validate this through controlled cross-domain protocols and human preference judgments.

D. Ethical Considerations

The current retrieval-grounded system is intended to augment user creativity rather than replace it. Risks include homogenization toward common exemplar neighborhoods and over-reliance on suggested settings. In future deployments, the interface should emphasize transparency (e.g., showing retrieved references) and user agency (e.g., direct editability and opt-out of retrieval). More generally, any model or retrieval corpus built from existing recordings or presets should consider copyright, licensing, and data provenance constraints.

VII. LIMITATIONS

Our approach has several limitations that matter directly for how the present evidence should be interpreted:

- **Scale and coverage.** The objective benchmark is a 204-query, guitar-only split with a single effect-chain topology. Resolved-audio grouping removes exact cross-split reuse, but cheap near-duplicate risk and database-density dependence remain. Larger multi-instrument and multi-effect benchmarks are required before making general retrieval-framework claims.
- **Metric semantics.** Normalized L2 improves comparability across heterogeneous parameter ranges, but it is not a perceptual normalization. Logarithmic, dB, and plugin-specific nonlinear controls can make equal normalized errors sound unequal, and the current metric-perception diagnostic does not validate the objective metrics as human-rating proxies.
- **Baseline boundary.** CLAP, PaSST, PANNs, and same-backbone ablations strengthen the retrieval comparison, but the range-projected MLP is stronger on normalized L2 and Acc@0.1. TRR should therefore be read as a retrieved-exemplar prior with editability and traceability benefits, not as a universally superior parameter estimator.
- **Listening.** The study lacks device metadata, participant-expertise labels, a documented manual-baseline time budget, and query-aligned ratings for the Protocol-A retrieval rows. It is exploratory perceptual context rather than a

fairness-critical human-vs-system benchmark or a validation of the objective metrics.

- **System scope.** Personalization, memory, and adaptive fusion are not validated under the same standard as Protocol-A. Protocol-C shows that degraded-input robustness depends on modality weighting, but the adaptive margin over strong fixed weights is small and should remain diagnostic.
- **Deployment cost and database dependence.** TRR adds Wav2Vec2 feature extraction and $O(D^2 T_{\text{frm}})$ Gram construction after projection. Caching and approximate indexing can reduce this cost, but live deployment and out-of-distribution database expansion require separate evaluation.

These limitations do not invalidate the present benchmark results, but they do define the boundary conditions under which our claims should be read. The current evidence is strongest for the narrower statement that texture-aware second-order structure can provide a useful retrieval prior for editable preset selection on this benchmark. Protocol-C further shows that degradation robustness is sensitive to fusion design, but it does not establish adaptive fusion as a generally superior multimodal method beyond the tested setting. Broader claims about universal similarity metrics, large-scale personalization, or robust deployment still require additional evidence.

VIII. CONCLUSION

We presented a retrieval-grounded approach to editable guitar-effect preset selection and identified Texture Resonance Retrieval (TRR) as the strongest retrieval representation in the reported P0 comparisons. Across objective metrics in Protocol-A (Table III), TRR improves parameter alignment over Wav2Vec, FeatureNN, CLAP, PaSST, and PANNs baselines, and we interpret this advantage as evidence for a more useful texture-aware retrieval prior rather than for a universal distance metric or a closed-form solution to inverse parameter estimation.

Taken together, the P0 evidence supports a narrow but practically relevant conclusion: retrieval over executable presets can provide editable starting points for DAW workflows, and second-order texture structure improves retrieval over selected audio embedding baselines on the audited guitar-effect preset benchmark. The degradation, listening, local-LLM, and supervised-regression diagnostics provide complementary boundary evidence, but they are not used to claim general system superiority.

Looking forward, the most important next steps are to evaluate executable direct parameter-regression and reranking baselines, expand the degradation and listening protocols, and test whether the same retrieval and fusion advantages transfer beyond guitar effects to broader production parameter spaces such as mixing and mastering workflows.

Overall, the present evidence supports texture-aware retrieval as a promising benchmark-backed foundation for editable guitar-effect preset selection, while leaving broader claims about universality, robustness, and personalization to future work.

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